

SovereignX Maintenance Archives

Historical Failure Incident Report - Pump P-204

Incident Date: 2025-05-18

Report Reference: INC-20250518-P204

Summary:

On May 18, 2025, Pump P-204 experienced an unexpected high-temperature shutdown event. The field operator logged bearing housing temperatures peaking at 92 C. This exceeded the SOP standard operation limit of 80 C. Simultaneously, radial vibration readings were recorded at 5.5 mm/s, well above the 4.0 mm/s standard permissible threshold.

Analysis:

The post-failure tear-down inspection revealed that the primary bearing housing suffered severe lubrication failure due to contaminants clogging the oil inlet. The lack of lubrication caused excessive friction, heat, and metal-to-metal wear, leading to elevated vibration.

Corrective Action:

The technician replaced the damaged bearings, flushed the oil chamber, and refilled the reservoir with clean lubricant. Follow-up testing confirmed normal temperatures (73 C) and low vibration (2.8 mm/s).

Preventive Recommendation:

Implement bi-monthly lubrication inspections and keep filter checks updated.